Angle-Aware Conformal Certification for Oriented Object Detection: Out-of-Sample Localization Coverage and Certified Recall on DOTA and DIOR-R

Zeyu Fu

***Abstract*—Oriented (rotated) object detectors ship without a distribution-free reliability guarantee. Porting conformal prediction to oriented boxes is not mechanical: the box angle is periodic with a $\pm 90^{\circ}$ seam discontinuity and degenerate at square boxes, so a naive coordinate-wise interval breaks. We conformalize a seam-continuous, square-safe Gaussian–Wasserstein (GWD) nonconformity score and build two distribution-free guarantees: (G1) per-detection localization coverage given detection, measured out-of-sample over $R{=}20$ scene splits (an in-sample audit we disclose and correct certifies nothing); and (G2) a Learn-then-Test (Hoeffding–Bentkus) certified rotated-IoU false-negative-rate bound that refuses below its power floor. Across RTMDet-R-l and Oriented R-CNN on DOTA-v1.0 and DIOR-R, G1 holds at nominal on both DIOR-R cells and ≈ 1–1.2 points below on both DOTA cells (a crop artifact); per-class Mondrian certificates refuse none ($15/15$ DOTA; $20/20$ DIOR-R), and G2 certifies $2/15$ DOTA and 5–$6/20$ DIOR-R classes, scaling with images-per-class. The preregistered Holm-8 set-size family (GWD smaller in all 8 cells at nominal α) is descriptive (degenerate for inference); a post-hoc coverage-matched ablation shows most of that gap is baseline over-coverage, the residual naive-baseline advantage concentrating on elongated objects within each frozen cell, not transferring across datasets (a registered HRSC2016-MS elongation prediction reversed on the added arm). A Config-A extension, further broadened by a Config-B cross-family DIOR-R breadth arm (RoI Transformer, S2A-Net), takes the study to ten G1 cells over three datasets and five detector architectures, refusing where support is thin. Ships as `rotcert`; the AOPG-table DIOR-R mAP reproduction is executed and disclosed, not claimed as passing.**

***Index Terms*—Oriented object detection, conformal prediction, uncertainty quantification, remote sensing, distribution-free coverage, rotated bounding boxes, DOTA, DIOR-R.**

I. Introduction

ORIENTED bounding boxes (OBBs) are the standard output format for detection in aerial, satellite, and document imagery, where objects appear at arbitrary in-plane rotations. Modern oriented detectors (RTMDet-R (1), Oriented R-CNN (2), LSKNet (3)) achieve strong accuracy on benchmarks such as DOTA (4) and DIOR-R (5), but a deployed detector emits a point prediction with no distribution-free statement about how far the true box may lie from it, nor about how many objects it silently misses.

Conformal prediction supplies exactly such guarantees, and conformal object detection is an established line: coordinate-wise/corner-wise interval constructions and conformal risk control for axis-aligned detectors (6, 7, 8, 9, 10), including one aerial (axis-aligned) treatment (11). The oriented case is not a mechanical port. An OBB is (cx, cy, w, h, θ) with a long-edge angle θ living on the quotient $\mathbb{R}/\pi\mathbb{Z}$: it is periodic, with a numerical discontinuity at the $\pm 90^{\circ}$ seam (a true axis at 89° and a prediction at -89° differ physically by 2° but numerically by 178°), and it is unidentifiable at square aspect ratios ($w = h$). A coordinate-wise conformal interval on θ neither wraps nor stays defined at squares, so it is structurally mis-specified in those regimes. We treat this as a structural argument and put it to an empirical, regime-conditional test in §V rather than asserting the failure as established: we find the naive baseline's coverage is indeed regime-dependent, though (over-covering globally via Bonferroni) it dips below nominal only in some cells — an honestly scoped result rather than a universal "naive breaks" claim.

Contribution. We conformalize a nonconformity score that is continuous across the angle seam and safe at square boxes — the 2-Wasserstein distance between the Gaussian representations of predicted and true boxes (the GWD, drawn from the oriented-detection loss literature (12, 13) but here used as a conformal score, not a loss). On this score we build a two-guarantee certification system: G1, per-detection localization coverage regions with a distribution-free marginal guarantee (14), measured out-of-sample; and G2, a Learn-then-Test (15) Hoeffding–Bentkus certified rotated-IoU false-negative-rate bound with an explicit power floor below which the tool refuses. We report a completed DOTA-v1.0 pilot, a multi-dataset DIOR-R extension (two in-house-trained detectors), a Config-A extension adding an HRSC2016-MS ship arm and a DOTA detector-zoo arm, and a Config-B extension adding cross-family detector breadth to DIOR-R itself (RoI Transformer (16), S2A-Net (17)) — ten G1 cells over three datasets and five detector architectures in all (§V-F, §V-G). Our claimed novelty is narrow and explicit: (i) the angle-aware OBB conformal score and the regime-conditional diagnosis of naive-conformal behaviour; (ii) the two-guarantee oriented-detection certification system; (iii) the `rotcert` tool. We do not claim conformal detection, conformal risk control / Learn-then-Test, the Gaussian OBB representation, or the detectors themselves — these are consumed. Figure 1 summarizes the resulting pipeline:

TODO-USER (affiliation): Z. Fu is with [Department, Institution, City, Country]. Correspondence: fuzeyu99@126.com.

TODO-USER (dates/funding): Manuscript prepared August 9, 2026; add received/revised dates and any funding acknowledgement at submission.

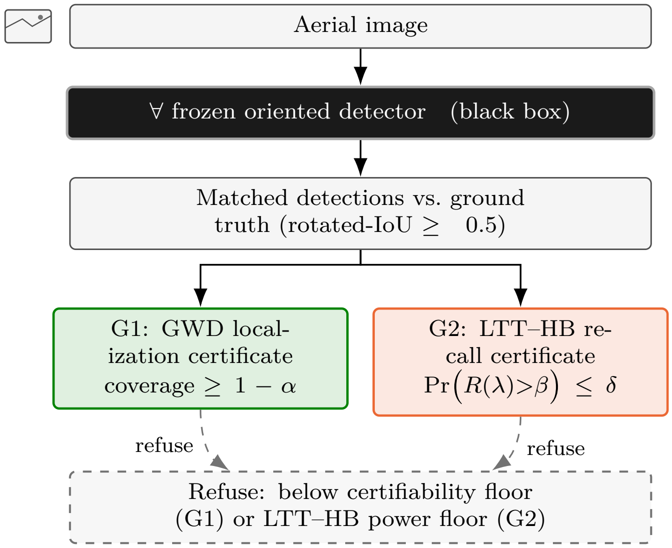

Fig. 1. Method overview. The wrapper treats any frozen oriented detector as a black box (five architectures certified in this paper: RTMDet-R-l, Oriented R-CNN, RoI Transformer, S2A-Net, plus the Config-A DOTA detector-zoo arm). Matched detections feed two independent, coupled certificates: G1 Mondrian (per-class) GWD calibration, giving a certified GWD-ball region at coverage $\geq 1-\alpha$; and G2, a Learn-then-Test Hoeffding–Bentkus (LTT–HB) certified rotated-IoU recall bound. Either branch refuses — rather than reports — on strata that do not clear its own certifiability floor ($\alpha_{\min} = 1/(n_{\text{cal}}+1) > \alpha$ for G1; below the LTT–HB power floor for G2).

an arbitrary frozen oriented detector is wrapped, never modified, and its matched detections feed two independent certificates — G1 (Mondrian GWD calibration) and G2 (Learn-then-Test certified recall) — either of which refuses on strata below its own certifiability floor rather than reporting a number it cannot back.

II. Related Work

Conformal and risk-controlled object detection. Split conformal prediction (14) and its Mondrian (group-conditional) variant (18) give finite-sample marginal coverage under exchangeability; adaptive, heteroscedasticity-aware conformal regression (19) is the broader motivation for conditioning on an informative stratum (class, here) rather than reporting one constant-width interval. For detection specifically, coordinate/corner-wise interval constructions and two-step uncertainty calibration (6, 7, 9), and conformal risk control / sequential risk control for recall-type risks (10, 8), target axis-aligned boxes. Learn-then-Test (15) provides high-probability risk certificates via multiple testing over a threshold grid, which we adopt for G2 – itself a multiple-testing instance of the more general risk-controlling-prediction-sets framework (20). Distribution-free structured-output UQ has also been built for other per-location targets, notably pixel-wise image-to-image regression (21), but not, to our knowledge, for the periodic, seam-discontinuous angle parameter of an oriented box. The one aerial treatment we are aware of (11) conformalizes an axis-aligned YOLO detector.

Oriented detection with conformal guarantees (the closest neighbour). The most adjacent system, EAV-DETR (22), couples arbitrary-view oriented UAV detection with a “Pose-Aware Mondrian Conformal Prediction” (PA-MCP) module that uses the UAV flight pose as the Mondrian stratum to produce conditional-coverage prediction sets. Its contribution is a single, bespoke high-accuracy detector: EAV-DETR is a real-time oriented-DETR variant whose gains come from two trained architectural components (scale-adaptive center supervision and anisotropic decoupled rotational attention), evaluated on two UAV benchmarks (CODrone, UAV-ROD), with the conformal layer calibrated on top of that detector and conditioned on a platform-specific pose prior. Our contribution is orthogonal and complementary: a detector-agnostic, purely post-hoc certification wrapper that treats any frozen detector as a black box — five detector architectures across three satellite/aerial benchmarks — strata on object class rather than on capture pose (so it needs no platform metadata), and adds machinery a single built-in guarantee does not carry: explicit certifiability-floor refusals, a coupled certified-recall guarantee via Learn-then-Test (15), and out-of-sample coverage validation. The two lines therefore complement rather than compete — a purpose-built UAV detector with an integrated pose-conditioned set, versus a reusable geometry-certification layer for the frozen detectors practitioners already deploy. (The differentiation here rests on EAV-DETR’s public abstract and code release; the full text is paywalled.)

Uncertainty quantification in the remote-sensing venues. Uncertainty-aware and evidential models are by now an established genre in the remote-sensing literature — e.g. uncertainty-aware building extraction (23), evidential open-set hyperspectral classification (24), and Bayesian uncertainty for explainable SAR target detection (25) — but these report heuristic or Bayesian uncertainty scores, not finite-sample coverage guarantees. We instead bring distribution-free, finite-sample conformal and risk-control guarantees to oriented detection in that setting.

Calibration of object detectors. A separate, longer-standing line recalibrates detector confidence scores post hoc, from classifier temperature scaling (26) to box-location/scale-conditional calibration for detection specifically (27) and calibration under domain shift (28). Calibration reduces the average miscalibration of a score but carries no finite-sample coverage guarantee for any individual stratum; G1 and G2 instead certify — and explicitly refuse below their respective certifiability/power floors rather than report a miscalibrated number.

Oriented detection and the angle representation. DOTA (4, 29) and DIOR-R (5) are the standard oriented benchmarks, with HRSC2016 (30) (used here as the elongation-dominated HRSC2016-MS arm, §V-F) and FAIR1M (31) the adjacent fine-grained oriented benchmarks we do not certify on. The five architectures our wrapper certifies span the major design lineages of oriented detection: two-stage RoI-warping (RoI Transformer (16), Oriented R-CNN (2)), single-stage feature alignment (S2A-Net (17), RTMDet-R (1)), and point-based representation (Oriented RepPoints (32), a Config-A zoo detector); the broader lineage these descend from includes the earliest rotated-anchor proposal networks (33), vertex-regression (34), progressive refinement (35), rotation-equivariant (36), and attention-based small/cluttered-object (37) designs — our wrapper is agnostic to which of these a practitioner deploys, treating each as a frozen black box. All in-house training uses the MMRotate toolbox (38). Gaussian-distance losses (GWD (12), KLD (13)) replace discontinuous angle regression with a smooth distance between box-induced Gaussians; we reuse that representation as a conformal score, not a loss.

III. Methods

The GWD nonconformity score. We canonicalize every ingested box to the `le90` long-edge convention ($w \geq h$, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2})$) and represent it as a 2-D Gaussian $\mathcal{N}(\mu, \Sigma)$ with $\mu = (cx, cy)$ and $\Sigma = R(\theta)\,\mathrm{diag}((w/2)^2, (h/2)^2)\,R(\theta)^\top$. Σ is a smooth function of (w, h, θ) with no seam discontinuity and no square degeneracy (as $w \to h$, Σ becomes isotropic and the ill-defined orientation is absorbed). The nonconformity between a prediction $\hat{p}$ and a ground-truth box g is the 2-Wasserstein (GWD) distance

$$s(\hat{p}, g)^2 = \|\hat{\mu} - \mu\|^2 + \mathrm{Tr}\Big(\hat{\Sigma} + \Sigma - 2\big(\hat{\Sigma}^{1/2}\Sigma\,\hat{\Sigma}^{1/2}\big)^{1/2}\Big),$$

with the closed-form 2×2 Bures term; an optional scale normalization $s/\sqrt{\hat{w}\hat{h}}$ is preregistered and ablated.

What the GWD-ball certifies (and what it does not). The G1 region is a ball in the GWD metric: $S(\hat{p}) = \{b : s(\hat{p}, b) \leq \hat{q}\}$ certifies, with probability $\geq 1 - \alpha$, that the true box lies within GWD distance $\hat{q}$ of the prediction. GWD is not rotated-IoU: distance in Gaussian-Wasserstein space is monotonically related to, but not a calibrated function of, box overlap, so the ball is not an IoU level-set and must not be read as "the true box is at IoU $\geq x$." The certified object is the GWD-ball itself; the per-parameter envelope we report for visualization (a bounding box on cx, cy, w, h and an arc on θ) is a conservative outer approximation and its per-axis widths are not calibrated marginal intervals. A user who needs an explicit IoU-coverage guarantee would conformalize a $1-$rotated-IoU score directly (an exploratory construction we also provide); GWD is chosen because it is simultaneously seam-continuous, square-safe, and closed-form, at the cost of this indirection.

G1 — per-detection coverage regions, measured out-of-sample. Given matched calibration pairs, let $\hat{q}$ be the $\lceil(1-\alpha)(n_{\mathrm{cal}}+1)\rceil/n_{\mathrm{cal}}$ empirical quantile of the scores. We calibrate per class (Mondrian, group = class) and refuse any stratum whose certifiability floor $\alpha_{\min} = 1/(n_{\mathrm{cal}}+1)$ exceeds α. Coverage is evaluated out-of-sample: over R=20 repeated scene-level splits (calibration 40% / matching 20% / held-out evaluation 40%, split-seed = repeat index), we fit $\hat{q}$ on the calibration scenes and measure coverage on the disjoint evaluation scenes, reporting the across-repeat mean and a BCa 95% CI.

G1 is conditional on detection. It certifies localization given a true positive at the operating rotated-IoU; missed detections and false positives are the job of G2.

G2 — certified rotated-IoU recall / FNR. Retaining detections with confidence $\geq \lambda$, define the per-image risk $R(\lambda)$ = mean miss rate, monotone in λ. We select λ by Learn-then-Test Hoeffding–Bentkus over a λ-grid controlling $\Pr(R(\lambda) > \beta) \leq \delta$. The exchangeable unit is the image, so per-class certification is limited by class-bearing images, not instances. Below the LTT-HB power floor ($n_{\mathrm{img}} \gtrsim \ln(G/\delta)/(2(\beta - \hat{R})^2)$, Bentkus-tightened) the tool refuses and prints the floor.

Baselines and the head-to-head. The same pipeline runs comparison scores through one interface: naive coordinate-wise CP with per-coordinate Bonferroni (B1, the canonical corner-wise conformal object-detection baseline (39)), axis-aligned-hull CP (B2), and wrapped-geodesic, doubled-angle, $1-$rotated-IoU ablations. We report the regime-conditional coverage of B1 in §V (descriptive). The preregistered efficiency head-to-head (class-blocked bootstrap, Holm-corrected across baseline $\times$ detector $\times$ dataset) ran under the signed freeze and is reported in §V-H, where GWD is smaller in all 8 of 8 cells but is reported as a preregistered descriptive outcome (the single-split test is degenerate; see the disclosure there).

Threats: an in-sample audit defect, found and corrected. An earlier pilot pipeline passed the same matched detection–ground-truth records to both the calibration and audit steps, so the coverage it reported was the split-conformal calibration-set coverage $\lceil(1-\alpha)(n+1)\rceil/n$ — for the DOTA pilot, 48,719/54,131 = 0.90002, which reproduces that in-sample identity exactly and the nominal target to ≈ 4 significant figures (0.9000). That number is an algebraic identity, independent of the score: any nonconformity function (GWD, naive, or noise) yields it, so it has zero power to validate GWD. We correct this by reporting the out-of-sample R=20 number throughout (§V); the in-sample value is retained only as a labelled self-consistency diagnostic, never as validation. This is disclosed here rather than left for a reader to infer.

Scene-level discipline. DOTA is tiled into overlapping 1024×1024 crops, so one object appears in adjacent crops; all splitting is at the level of the source scene (never the crop), and every coverage CI is scene-clustered. The out-of-sample R=20 protocol also probes the box-level exchangeability approximation: because each repeat is an independent scene-level partition, the spread of the 20 per-repeat coverages (reported as min/max) is a direct check that the point estimate is stable under scene re-partition, not merely that its CI is wide.

IV. Experimental Setup

DOTA-v1.0 pilot. We certify RTMDet-R-l (frozen Apache-2.0 MMRotate (38) zoo checkpoint) on the DOTA-v1.0 (4) validation split (15 classes); calibration and evaluation use val only (the test server is untouched). Boxes are canonicalized to `le90`; matching is greedy one-to-one at rotated-IoU $\geq$ 0.5 within class. Levels are $\alpha = 0.10$ (G1) and $(\beta, \delta) = (0.20, 0.05)$ (G2). G1 coverage is the out-of-sample R=20 scene-split number.

DIOR-R extension. No license-clean DIOR-R-trained checkpoint exists, so both detectors (Oriented R-CNN R-50, RTMDet-R-l) are trained in house on DIOR-R (5) trainval via Apache-2.0 mmrotate; certification runs on the DIOR-R test split (11,738 images; single-tile, scene = image) under the identical protocol. Training setup. Oriented R-CNN is trained 1× (12 epochs) with AdamW (lr = 10^{-4}, weight decay 0.05) rather than the config-default SGD, together with an annotation-filtering step that drops degenerate zero-area DIOR-R ground-truth boxes (removing boxes below a minimum width or height, and images left with no valid ground truth); both changes were made after those boxes destabilized the default recipe. RTMDet-R-l is trained $3\times$ (36 epochs). Checkpoints are saved at the 12-epoch interval with no best-checkpoint hook, so each detector is deployed at the final epoch of its fixed schedule. For RTMDet-R-l the intermediate epoch-24 checkpoint scored higher on the test split (69.65) than the deployed epoch-36 (68.36); because the validation loop evaluated on that same test split, selecting epoch 24 would be checkpoint selection on the evaluation set, so we deploy the final epoch and report the lower number rather than adopt the higher one.

mAP reproduction (AOPG gate, preregistered K3), executed post-freeze. The in-house detectors reproduce DIOR-R test-split mAP of 62.61 (Oriented R-CNN, $1\times$) and 68.36 (RTMDet-R-l, $3\times$), against the AOPG published DIOR-R reference of 64.41 (5). The AOPG table lists neither detector as a same-method row, so its ± 0.5 identity-reproduction tolerance cannot be met by construction (-1.80, $+3.95$ vs. AOPG). A same-method external anchor does exist: a recent benchmark table reports Oriented R-CNN R-50 at 64.30 mAP on DIOR-R (40) — essentially AOPG's 64.41 and within 1.7 points of our $1\times$ reproduction (62.61); the DIOR-R tables that state a schedule use 12 epochs ($1\times$), matching ours. We could not verify any same-method Oriented R-CNN figure near the 67–68 leaderboard band a naive comparison invites: that band is recent end-to-end transformer detectors, not Oriented R-CNN. Both reproduced scores fall in the published DIOR-R band (Oriented R-CNN between Gliding Vertex and RoI Transformer at $1\times$; RTMDet-R-l above the entire 2021 table). We disclose the gap rather than chase it, and certification uses reproduced scores only, never paper-quoted ones.

V. Results

A. G1 — out-of-sample localization coverage across the four original cells

The GWD certificate's held-out coverage is at nominal on both DIOR-R cells and $\approx$ 1–1.2 points below on both DOTA cells (Table I). Decisively, the full detector$\times$dataset cross now exists: both detector families cover at nominal on DIOR-R (0.9004/0.9012) and below on DOTA (0.8902/0.8879) — the shortfall appears to be a dataset artifact — DOTA's overlapping 1024-px crops leave residual within-scene correlation that scene-level splitting reduces but does not remove — not a detector or score defect. The in-sample audit (Table I, right column) is a labelled diagnostic only (see Methods/Threats): it equals 0.900 by construction for any score.

B. G1 — per-class Mondrian certificates (all classes certified, none refused)

The Mondrian calibration certifies every class with 0 refused in every cell: $15/15$ on DOTA and $20/20$ on DIOR-R for both Oriented R-CNN and RTMDet-R-l. Table II lists the DOTA pilot's per-class GWD-ball radii, spanning nearly two orders of magnitude (compact ships/vehicles tightest; elongated ground-track-field/harbor widest); the DIOR-R rosters show the same compact-vs-infrastructure spread (e.g. DIOR ship $\hat{q}$=3.99, area $50\,\mathrm{px}^2$; dam $\hat{q}$=73.9, area $17{,}134\,\mathrm{px}^2$).

TABLE I
G1 marginal coverage, out-of-sample R=20 scene-split (BCa 95% CI; nominal 0.90). Cells: DOTA pilot = RTMDet-R-l on DOTA-v1.0; DOTA-ORCNN = Oriented R-CNN on DOTA-v1.0; DIOR-ORCNN = Oriented R-CNN on DIOR-R; DIOR-RTMDet = RTMDet-R-l on DIOR-R. The in-sample column is the tautological self-consistency value ($\lceil(1-\alpha)(n+1)\rceil/n$), kept as a diagnostic only. From the frozen out-of-sample coverage records.

Cell	OOS R=20 coverage [BCa 95% CI]	in-sample (diag.)
DOTA pilot	0.8879 [0.8774, 0.8979]	0.900
DOTA-ORCNN	0.8902 [0.8780, 0.8995]	0.900
DIOR-ORCNN	0.9004 [0.8987, 0.9020]	0.900
DIOR-RTMDet	0.9012 [0.8995, 0.9030]	0.900

TABLE II
G1 per-class GWD certificates on the DOTA-v1.0 pilot ($\alpha = 0.10$). $\hat{q}$ is the GWD-ball radius; area is the (cx, cy) coverage-slice area $\pi\hat{q}^2$ (px^2). All 15 classes certified, 0 refused. From the frozen per-class certification record.

Class	n_{cal}	$\hat{q}$	Area (px^2)
ship	18,125	3.50	38.6
small-vehicle	10,394	3.14	30.9
large-vehicle	8,727	3.73	43.7
storage-tank	4,291	2.80	24.6
plane	4,299	7.48	175.6
harbor	4,007	19.14	1,151.0
tennis-court	1,510	2.97	27.8
bridge	681	9.34	274.3
swimming-pool	660	6.59	136.3
baseball-diamond	347	10.30	333.1
basketball-court	266	6.37	127.4
roundabout	262	5.55	96.8
soccer-ball-field	238	17.52	963.9
ground-track-field	205	22.09	1,533.1
helicopter	119	12.01	452.9

A recovered-classes provenance note (DIOR-R). The first DIOR-R Oriented R-CNN run certified $18/20$ classes: two multi-word classes (expressway-service-area and expressway-toll-station) were excluded because the detector dump lower-cased them while the ground truth capitalized them, so the class-exact matcher found 0 true positives — the tool correctly excluded-and-counted them. Re-running against a casing-normalized ground truth recovers both $(20/20)$ and leaves the other 18 classes' per-class $\hat{q}$ and n_{cal} bit-identical (verified class-by-class), and out-of-sample coverage unchanged ($0.9001 \to 0.9004$) — a surgical fix that also demonstrates the refusal machinery behaving as designed on a real label-hygiene fault.

Per-class conditional coverage on the 20-class dataset (post-hoc, descriptive). A hostile reviewer may ask whether the Mondrian certificate merely holds marginally while hiding per-class shortfalls. It does not. Mirroring the shipped Mondrian machinery (grouping on object class) under the same R=20 scene-split protocol, we measure the out-of-sample coverage of the per-class GWD certificate within each of the 20 DIOR-R classes (Table III). Every class clears the G1 certifiability floor

$\alpha_{\min}=1/(n_{\text{cal}}+1)<\alpha$ (0/20 refused; the smallest class, dam, has $n_{\text{cal}}=266$), and per-class conditional coverage is tightly clustered around nominal: pooled per-class coverage is 0.900 in both cells, and no class deviates more than ≈ 1.3 points from 0.90 (Oriented R-CNN range 0.887 golffield – 0.913 dam; RTMDet-R-l range 0.891 stadium – 0.907 tenniscourt), with the sub-nominal classes inside one split-to-split standard deviation of 0.90. The certifiability floor that does bite is the recall certificate's (G2) images-per-class power floor, not the localization certificate: the small classes it refuses (15/20 for Oriented R-CNN, 14/20 for RTMDet-R-l — last column) are still localization-certified at G1 with coverage at nominal. The geometry certificate thus covers every class approximately at nominal (within ≈ 1.3 points, one split-to-split std, of every sub-nominal class), while the recall certificate honestly abstains on the rare ones.

C. G2 — certified recall scales with images-per-class

At $(\beta, \delta) = (0.20, 0.05)$, G2 certifies a per-class FNR bound where powered and refuses elsewhere (Table IV, all nine cells). The certified count rises from 2/15 on DOTA ($\sim$ 458 val images) to 5/20 and 6/20 on DIOR-R (11,738 test images), and RTMDet-R-l (denser detections, more matched TPs) certifies one more class than Oriented R-CNN — direct evidence the LTT-HB power floor is data-scale driven and moves down with more images, not a permanent ceiling. Refusals are of two honest kinds: an infinite power floor (realized per-image miss rate $\geq \beta$, uncertifiable at any n; e.g. DIOR-ORCNN airplane, vehicle — RTMDet's floors for the same two classes are finite (1,225 and 75,149 images, $\hat{r} < \beta$), refused only for insufficient images) and insufficient images below the Bentkus floor (e.g. DOTA harbor $n_{\text{img}}=112 < 135.7$). The tool prints the floor rather than reporting an uncertifiable number.

D. Naive coordinate-wise CP under-covers conditionally, not universally

We put the paper's motivating hook to a direct regime-conditional test (Table V, descriptive). The naive coordinate-wise (Bonferroni-box) baseline over-covers marginally everywhere (0.933–0.936), as Bonferroni predicts, but its coverage is strongly regime-dependent: the seam (boundary) or square is always its most-depressed regime relative to interior. Whether it falls below nominal is cell-dependent — strongly on DOTA square (0.8405, CI $[0.801, 0.864]$ below nominal; an 11-point deficit vs. its 0.9505 interior), marginally on DIOR-ORCNN boundary (0.8976), and not at all on DIOR-RTMDet (boundary 0.9121, still above nominal). GWD is seam/square-stable on both DIOR-R cells and degrades least at square on the RTMDet-DOTA pilot ($\approx$ 0.88 vs. naive's 0.8405); on the DOTA-ORCNN cell, however, both scores show a comparable square-regime deficit (GWD 0.8612, naive 0.8591, in-sample diagnostic), so GWD's square-regime advantage is itself detector-dependent on DOTA rather than universal. We therefore state the hook as a conditional claim, and note that the preregistered set-size head-to-head is reported in §V-H (GWD smaller in 8/8 cells, reported descriptively); the regime-conditional under-coverage contrast remains descriptive here.

The Config-B DIOR-R extension is regime-consistent for RoI Transformer but not for S2A-Net. Repeating this audit on the two Config-B DIOR-R cells (§V-G) extends the table to all four DIOR-R detectors. RoI Transformer matches the established DIOR-R pattern (boundary 0.902, at nominal, not below it, like DIOR-RTMDet). S2A-Net does not: its boundary coverage (0.867) is the only DIOR-R regime cell in the study that dips clearly below nominal — deeper than DIOR-ORCNN's marginal dip (0.898) and unlike DIOR-RTMDet or RoI Transformer, both of which sit at or above nominal there — while its square over-coverage (0.988) is the highest of any cell in the table. The direction (boundary worst, square best) is unchanged, but the magnitude is not: the naive baseline's regime-dependence on DIOR-R is detector-dependent, and S2A-Net is the cell where it is most severe, closer in kind to the DOTA-square deficit than to the mild DIOR boundary dips seen so far.

E. τ sensitivity (matched-TP conditioning is not fragile)

Re-matching at rotated-IoU thresholds $\tau \in \{0.5, 0.6, 0.7\}$ (DIOR Oriented R-CNN) tightens the certificate: mean (cx, cy)-slice area $6{,}514 \to 4{,}409 \to 2{,}729\,\text{px}^2$ (over the 18 classes powered at all three τ) as τ rises, because higher-IoU true positives have smaller localization error. The realized coverage stays pinned at nominal (0.900) across all three τ, but that number is the in-sample calibration-set coverage and is tautologically $\approx$nominal by the split-conformal construction, so it is a consistency check, not independent evidence. The matched-TP conditioning is thus robust to the match threshold.

F. Config-A extension: canonical trio and detector-zoo breadth

Four further G1 cells under the identical frozen pipeline ($\alpha = 0.10$, GWD, R=20 scene splits) extend the study from two datasets to three and to four detector architectures: the elongation-dominated HRSC2016-MS ship benchmark (30) (multi-scale variant; two in-house-trained detectors, Oriented R-CNN 1$\times$ and RTMDet-R 3$\times$, mirroring the DIOR-R configs) and a DOTA-v1.0 detector zoo (RoI Transformer, Oriented RepPoints (32)). The GWD certificate holds at nominal by point estimate in every new cell and the naive baseline over-covers throughout (Table VI), though the low-support RoI Transformer cell (70 scenes) carries a wide audit CI ($[0.823, 0.945]$).

Refusals, exclusions, and a reversed prediction. G2 recall refuses in all four new cells at $(\beta, \delta) = (0.20, 0.05)$ — the LTT-HB power floor is infinite at these image counts (HRSC $n_{\text{img}} = 440$, $\hat{r} = 0.5464/0.6076$; DOTA zoo $n_{\text{img}} = 1093$, $\hat{r} = 0.9380/0.8496$) — the refusal mechanism working as designed, not a failure. Two tiny Oriented RepPoints strata are G1-refused by the certifiability floor $\alpha_{\min} = 1/(n_{\text{cal}} + 1)$ (large-vehicle $n_{\text{cal}} = 3$, soccer-ball-field $n_{\text{cal}} = 4$); we planned six DOTA zoo detectors and realized four, excluding S2A-Net (17) and Gliding Vertex (34) (mmrotate v0.1.0 checkpoints incompatible with dev-1.x). Finally, the registered HRSC elongation prediction (logged pre-run: HRSC pooled coverage-matched ratio versus naive should reach low-0.7$\times$) is reversed: realized 1.2413 (Oriented R-CNN) and 1.0198 (RTMDet-R), both ≥ 1.0, with the zoo adding detector-dependence in the

TABLE III

PER-CLASS OUT-OF-SAMPLE CONDITIONAL COVERAGE OF THE GWD CERTIFICATE ON THE TWO CORE DIOR-R CELLS ($\alpha = 0.10$, NOMINAL 0.90; R=20 SCENE SPLITS, MEAN OVER REPEATS). n_{cal} IS THE ORIENTED R-CNN FULL-DATA PER-CLASS CALIBRATION COUNT (RTMDET-R-L COUNTS ARE WITHIN A FEW PERCENT). G2 RECALL: O=CERTIFIED FOR ORIENTED R-CNN, R=CERTIFIED FOR RTMDET-R-L, —=REFUSED BY THE LTT-HB POWER FLOOR. POST-HOC / DESCRIPTIVE; FROM THE FROZEN PER-CLASS CONDITIONAL-COVERAGE RECORD.

Class	n_{cal}	ORCNN cov.	RTMDet cov.	G2 recall
ship	30,731	0.899	0.899	—
storagetank	16,081	0.900	0.900	R
vehicle	12,958	0.904	0.899	—
tenniscourt	6,498	0.897	0.907	O,R
airplane	5,589	0.898	0.900	—
baseballfield	2,617	0.901	0.898	R
windmill	2,291	0.900	0.895	—
basketballcourt	1,928	0.898	0.894	O,R
groundtrackfield	1,773	0.903	0.905	O,R
harbor	1,692	0.901	0.899	—
bridge	1,470	0.906	0.902	—
overpass	1,183	0.898	0.895	—
expressway-service-area	962	0.896	0.905	O,R
chimney	808	0.899	0.900	—
stadium	600	0.906	0.891	O
expressway-toll-station	508	0.913	0.898	—
golffield	459	0.887	0.899	—
airport	357	0.906	0.896	—
trainstation	354	0.902	0.899	—
dam	266	0.913	0.903	—

TABLE IV

G2 CERTIFIED-RECALL COUNTS ACROSS ALL NINE CELLS (β=0.20, δ=0.05). CERTIFIED COUNT SCALES WITH IMAGES-PER-CLASS: THE HIGH-IMAGE DIOR-R CELLS CERTIFY 5–7/20, WHILE THE LOW-IMAGE HRSC2016-MS (n_{img}=440) AND DOTA-V1.0 ZOO (n_{img}=1093) CELLS REFUSE AT THE LTT-HB POWER FLOOR. FROM THE PER-CELL CERTIFICATION OUTPUTS.

Cell	cert. / total	certified classes ($\lambda^\star$, realized risk), or refusal
DOTA pilot	2/15	small-vehicle (0.051, 0.024), tennis-court (0.094, 0.001)
DIOR-R ORCNN	5/20	basketballcourt, tenniscourt, stadium, groundtrackfield, expressway-service-area
DIOR-R RTMDet-R-l	6/20	basketballcourt, tenniscourt, groundtrackfield, expressway-service-area, baseballfield, storagetank
DIOR-R RoI Transf.	5/20	basketballcourt, tenniscourt, stadium, groundtrackfield, expressway-service-area
DIOR-R S2A-Net	7/20	baseballfield, basketballcourt, groundtrackfield, ship, stadium, storagetank, tenniscourt
HRSC ORCNN	0/1	refused: LTT-HB power floor (n_{img}=440, $\hat{r}$=0.5464)
HRSC RTMDet-R	0/1	refused: power floor (n_{img}=440, $\hat{r}$=0.6076)
DOTA RoI Transf.	0/15	refused: power floor (n_{img}=1093, $\hat{r}$=0.9380)
DOTA RepPoints	0/15	refused: power floor (n_{img}=1093, $\hat{r}$=0.8496)

TABLE V

ANGLE-REGIME COVERAGE OF THE NAIVE COORDINATE-WISE BASELINE VS. GWD (IN-SAMPLE AUDIT, NOMINAL 0.90; DESCRIPTIVE). NAIVE OVER-COVERS MARGINALLY BUT DIPS BELOW NOMINAL ONLY IN SOME CELLS. EXTENDED TO ALL FOUR DIOR-R CELLS (ROI TRANSFORMER, S2A-NET ADDED, §V-G). FROM THE FROZEN REGIME-AUDIT RECORDS.

Cell	naive marg.	naive interior	naive boundary	naive square	GWD (int/bnd/sq)
DOTA pilot	0.933	0.951	0.919	0.841	0.893/0.937/0.880
DIOR-ORCNN	0.935	0.944	0.898	0.962	0.886/0.910/0.933
DIOR-RTMDet	0.935	0.945	0.912	0.940	0.885/0.917/0.927
DIOR-RoI-Trans	0.936	0.945	0.902	0.960	0.885/0.910/0.934
DIOR-S2ANet	0.935	0.944	0.867	0.988	0.884/0.911/0.951

same direction (1.6618 RoI Transformer vs. 0.7056 Oriented RepPoints). The elongation mechanism therefore does not transfer across datasets or detectors; what survives is the within-cell pattern on the four original cells (descriptive, post-hoc), and GWD never beats the axis-aligned hull (new cells 1.30–2.44×). Population caveats: HRSC2016-MS has essentially no compact stratum, and the zoo subset is markedly less elongated than the frozen DOTA cells, so the zoo ratios are detector-breadth evidence, not a third elongation point.

TABLE VI

EXTENSION G1 CELLS ($\alpha = 0.10$, NOMINAL 0.90), ALL UNDER THE SAME FROZEN PIPELINE AS THE CORE CELLS (TABLE I). AUDITED MARGINAL COVERAGE (GWD POINT [CI] AND NAIVE-COORD POINT), THE OUT-OF-SAMPLE R=20 RECALIBRATION MEAN $[\min, \max]$, THE MATCHED TRUE-POSITIVE COUNT n_{TP} AND THE AUDIT SCENE COUNT. CONFIG-A (§V-F) ADDS HRSC2016-MS AND A DOTA-V1.0 DETECTOR ZOO (THE LOW-SUPPORT ROI TRANSFORMER CELL HAS ONLY 70 SCENES, HENCE ITS WIDE GWD CI); CONFIG-B (§V-G) ADDS TWO CROSS-FAMILY DIOR-R CELLS. FROM THE FROZEN CONFIG-A AND CONFIG-B CERTIFICATION RECORDS.

Detector · dataset	GWD cov. [CI]	naive cov.	R=20 mean $[\min, \max]$	n_{TP}	scenes
Config-A — HRSC2016-MS and DOTA-v1.0 detector zoo (§V-F)					
Oriented R-CNN · HRSC2016-MS	0.9011 $[0.8798, 0.9201]$	0.9212	0.8989 $[0.8528, 0.9500]$	799	314
RTMDet-R · HRSC2016-MS	0.9015 $[0.8760, 0.9227]$	0.9179	0.9085 $[0.8659, 0.9568]$	670	310
RoI Transformer · DOTA-v1.0	0.9004 $[0.8232, 0.9447]$	0.9279	0.8882 $[0.7431, 0.9736]$	4,246	70
Oriented RepPoints · DOTA-v1.0	0.9003 $[0.8545, 0.9299]$	0.9329	0.8988 $[0.7818, 0.9720]$	5,617	205
Config-B — cross-family DIOR-R (§V-G)					
RoI Transformer · DIOR-R	0.9000 $[0.8961, 0.9036]$	0.9364	0.9014 $[0.8919, 0.9113]$	89,974	10,966
S2A-Net · DIOR-R	0.9000 $[0.8959, 0.9041]$	0.9349	0.9006 $[0.8941, 0.9074]$	95,192	10,940

G. Config-B extension: cross-family detector breadth on DIOR-R

Two further G1 cells add cross-family detector breadth to DIOR-R itself, under the identical frozen pipeline as the two original DIOR-R cells (§IV): RoI Transformer (two-stage) and S2A-Net (single-stage), both trained in house $1\times$ on DIOR-R trainval and certified on the same 11,738-image test split. (S2A-Net's native training convention is `le135`; this has no bearing on the certified score, since every emitted box is re-canonicalized to `le90` on output regardless of the model's internal angle convention, §III.) The GWD certificate again holds at nominal on both (audited 0.9000/0.9000; R=20 mean 0.9014/0.9006), and naive over-covers as usual (0.9364/0.9349 audited), matching the two original DIOR-R cells (Table VI). Both new cells certify every Mondrian stratum at G1 ($20/20$, 0 refused for GWD and naive-coord), extending the "zero G1 refusals on DIOR-R" record to all four detector×DIOR-R cells.

G2 recall and per-class conditional coverage. At $(\beta, \delta) = (0.20, 0.05)$, G2 certifies $5/20$ classes for RoI Transformer — the identical roster the Oriented R-CNN cell certifies (basketballcourt, tenniscourt, stadium, groundtrackfield, expressway-service-area) — and $7/20$ for S2A-Net (baseballfield, basketballcourt, groundtrackfield, ship, stadium, storagetank, tenniscourt), the highest certified count of any DIOR-R cell, consistent with S2A-Net's matched-TP population (95,192, the densest of any cell) driving the LTT-HB power floor down further; both new cells' pooled recall refuses only at the power floor ($n_{\mathrm{img}} = 19{,}435$, $\hat{r} = 0.2230/0.2118$), the same refusal mechanism operating as designed elsewhere (§V-F). Repeating the per-class conditional-coverage analysis of §V (Table III) on these two cells (now merged into a single four-cell per-class conditional-coverage record) shows the same pattern holds: $0/20$ classes refused at the G1 certifiability floor in either cell, and out-of-sample per-class coverage stays inside the paper's existing ≈ 1.3-point band around nominal (RoI Transformer 0.891 groundtrackfield – 0.910 dam; S2A-Net 0.889 airport – 0.912 trainstation) — the conditional-coverage finding is not an artifact of the two original detectors. Figure 2 plots the full four-cell per-class record (all 20 classes across the four DIOR-R detectors), visualizing the tight band around nominal reported above.

Training-seed stability (HRSC 3-seed arm; post-hoc). As a post-hoc, exploratory Config-B arm we retrained the two in-house HRSC detectors under seeds $1, 2$ (seed 0 is the Config-A cell, not recomputed) and re-ran the frozen certification (Table VII). Coverage lands at nominal for every seed — as split-conformal calibration guarantees under exchangeability, not as a nontrivial stability finding: the audited GWD coverage sits within 0.0013 of a common value (Oriented R-CNN 0.9011–0.9024; RTMDet-R 0.9015–0.9023), i.e. no seed-driven exchangeability breakdown occurred, and split conformal pins the marginal coverage by construction. The seed-sensitive quantity is instead the region scale: the conformal quantile $\hat{q}$ (the GWD nonconformity radius) moves only 1.5% across seeds for Oriented R-CNN (39.57–40.17) but $\approx 12\%$ for RTMDet-R (48.23–54.26), so it is the certified region's size, not its coverage, that carries the training-seed variation. The six R=20 coverage means all straddle nominal (0.8903–0.9113), with RTMDet-R's wider across-seed spread (0.021) inside each cell's own reshuffle range ($[\min, \max]$ spans $\approx$ 0.06–0.13), and the G2 power-floor refusal ($n_{\mathrm{img}} = 440$) fires identically in all six cells. Matched-true-positive counts vary with seed (630–799) as expected detector-side variation.

H. Preregistered head-to-head: the Holm-8 family (demoted to descriptive)

Preregistered result (descriptive). The preregistered family — 2 baseline contrasts (naive coordinate-wise + Bonferroni; axis-aligned hull) × 2 detectors × 2 datasets = 8 one-sided tests (H1: GWD region strictly smaller) — was executed exactly as frozen under the signed freeze on a single frozen $40/20/40$ cal/match/eval scene split (split-seed 0) per cell; the R=20 across-split robustness below is a separate, exploratory analysis. GWD is strictly smaller in all 8 of 8 cells. We report this as a preregistered descriptive outcome, not statistical confirmation: as detailed next the single-split test is structurally degenerate, so the nominal Holm-adjusted $p = 0.004$ is an artifact of the $1/2001$ permutation floor and carries no evidence strength. The eval-split region-size ratios (GWD ÷ baseline; "region size" here = the (cx, cy)-slice area, $\pi\hat{q}^2$ for GWD and $4\hat{q}_{cx}\hat{q}_{cy}$ for the union-bound baselines — a coverage-region slice, not a calibrated marginal interval; see §III) of 0.67/0.86 (RTMDet–

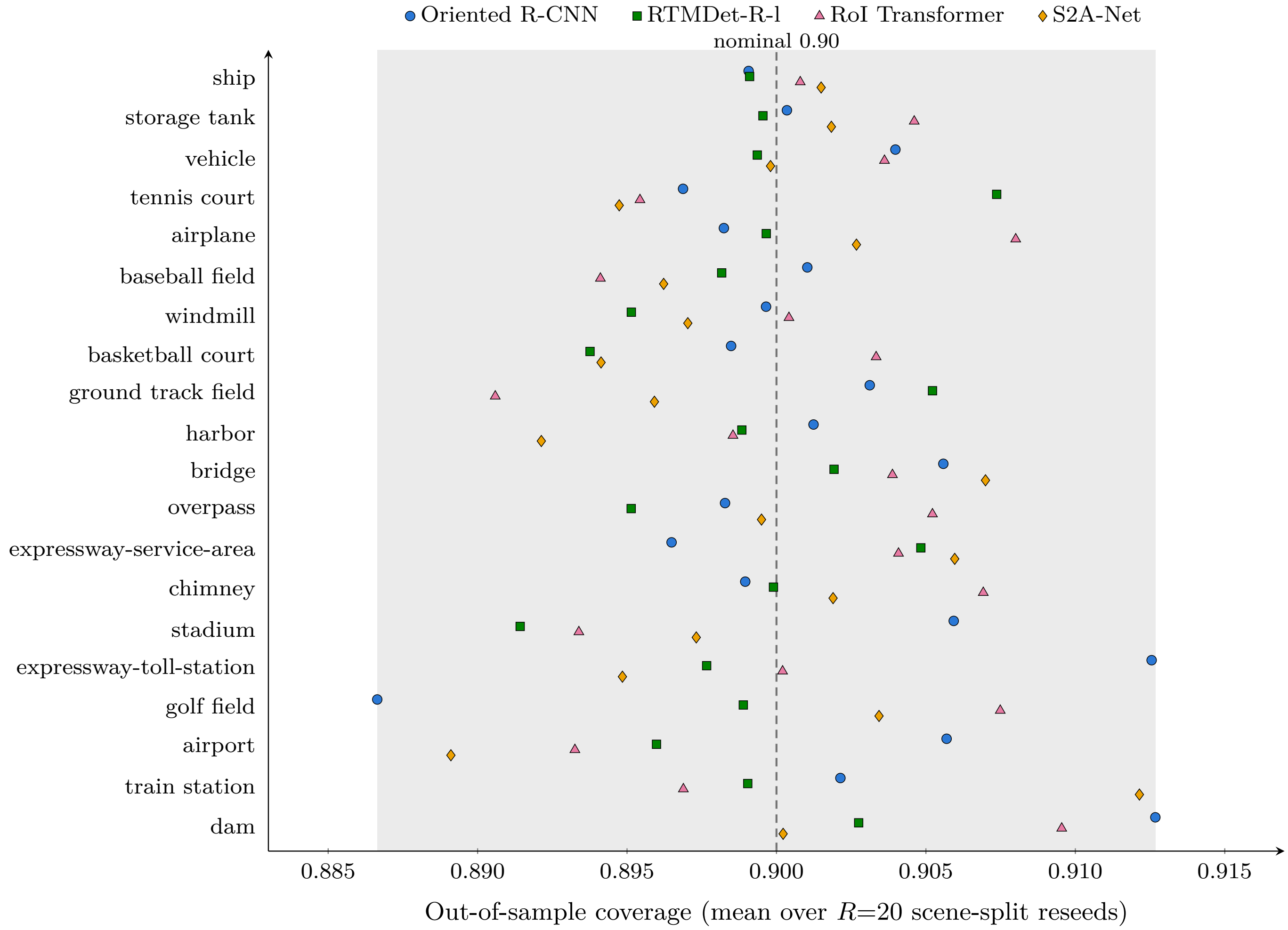

Fig. 2. Per-class out-of-sample conditional coverage of the GWD certificate on all four DIOR-R detector cells ($\alpha = 0.10$, nominal 0.90, mean over R=20 scene-split reseeds). Each of the 80 class×cell points falls within the shaded ≈ 1.3-point band around the dashed nominal line. From the frozen four-cell per-class conditional-coverage record (merges Table III's two cells with the Config-B RoI Transformer and S2A-Net cells, §V-G).

TABLE VII
HRSC 3-SEED ARM (POST-HOC/EXPLORATORY CONFIG-B): PER-SEED AUDITED GWD COVERAGE (POINT [CI]), MATCHED TRUE-POSITIVE COUNT, AND OUT-OF-SAMPLE R=20 RECALIBRATION MEAN [min, max] AT $\alpha = 0.10$ (NOMINAL 0.90), SAME FROZEN PIPELINE. SEED 0 IS THE CONFIG-A CELL (NOT RECOMPUTED); G2 REFUSES (POWER FLOOR) IN ALL SIX CELLS.

Detector	seed	GWD cov. [CI]	n_{TP}	R=20 mean [min, max]
Oriented R-CNN	0	0.9011 [0.8798, 0.9201]	799	0.8989 [0.8528, 0.9500]
Oriented R-CNN	1	0.9023 [0.8803, 0.9228]	727	0.8982 [0.8414, 0.9360]
Oriented R-CNN	2	0.9024 [0.8780, 0.9246]	758	0.9005 [0.8249, 0.9545]
RTMDet-R	0	0.9015 [0.8760, 0.9227]	670	0.9085 [0.8659, 0.9568]
RTMDet-R	1	0.9016 [0.8799, 0.9235]	630	0.9113 [0.8839, 0.9402]
RTMDet-R	2	0.9023 [0.8796, 0.9232]	737	0.8903 [0.8444, 0.9311]

DOTA, vs. naive/hull), 0.60/0.78 (ORCNN–DOTA), 0.33/0.48 (ORCNN–DIOR-R), and 0.40/0.59 (RTMDet–DIOR-R): GWD is strictly smaller in every cell.

Degeneracy disclosure — the p-values carry no evidence strength. Under the frozen protocol both constructions are calibrated with pooled split-conformal (one marginal cell, no Mondrian grouping) — not the per-class Mondrian calibration of the shipped G1 certificate (§V), a deliberate mismatch we flag here. (The contrast also pooled all matched true positives without excluding square-stratum detections; the design's square-vacuity rule guards the angular set, not this center-localization (cx, cy) slice, so the effect on the area metric here is limited.) Pooled calibration makes each construction's region-size a single scalar per cell: $\pi\hat{q}^2$ and $4\hat{q}_{cx}\hat{q}_{cy}$ are functions of scalar quantiles only, so every detection in a cell carries the identical set size (we re-verified 1 unique value over all 20k–40k eval pairs per cell; e.g. ORCNN–DOTA GWD 126.6, naive 212.5, hull 162.3 px^2). The preregistered per-pair machinery is therefore degenerate: every per-pair log-ratio equals the same negative constant, so the class-blocked bootstrap CI is zero-width and the sign-flip permutation p is deterministic at the $1/(n_{\text{perm}}+1) = 1/2001$ floor for any negative effect, independent of magnitude; the effective sample size is 1 (a pair of scalars from one split), not the 20k–40k pairs. The preregistered test thus certifies only the per-split deterministic inequality $\pi\hat{q}^2_{\text{GWD}} < 4\hat{q}_{cx}\hat{q}_{cy}$ on the frozen split; its substantive content is the effect size (the ratios above), not the p-values, and "$8/8$ at the p-floor" is not eight independent strong rejections. This inequality is moreover close to analytic: the union-bound (Bonferroni) baselines over-cover by construction at any fixed nominal α, so at matched nominal level their (cx, cy) box is predictably larger than a near-nominal ball — the "$8/8$ smaller" direction largely reflects that structural Bonferroni looseness, not a data-driven margin. The coverage-matched ablation below is precisely what removes it.

Across-split sensitivity of the nominal-level contrast (R=20; not preregistered). We recompute the per-split GWD and baseline region sizes on the same R=20 scene splits as the G1 coverage headline (split-seed = repeat index), to check how the frozen single-split nominal-level ratio varies by realization. The direction is stable: GWD's region is smaller on all 20 of 20 splits in every one of the 8 cells, no split's ratio crosses 1, and the worst single split anywhere is 0.974 (RTMDet–DOTA vs. hull). Table VIII reports per-cell mean and $[\min, \max]$ ratios. We attach no p-value to this $20/20$ count: the split procedure reshuffles one fixed scene set, so the 20 splits are dependent, overlapping partitions, not 20 independent draws — an exact sign test would treat them as independent and materially overstate the evidence. We therefore read this only as a split-sensitivity range of the (still descriptive) frozen nominal-level contrast, which it corroborates and nowhere contradicts. Crucially, this analysis (like the frozen family) compares at matched nominal α, where the baselines over-cover; the coverage-matched ablation next is what tests the efficiency claim on a coverage-fair footing, and is our primary quantitative support for it.

Coverage-matched ablation — the efficiency claim on a coverage-fair footing. The contrasts above equalize the nominal $\alpha = 0.10$, not realized coverage. At that level GWD attains close-to-nominal eval coverage (R=20 mean 0.883 on DOTA and 0.899 on DIOR-R, consistent with the G1 headline), while the union-bound baselines over-cover — R=20 mean realized coverage 0.915–0.948 across cells — so part of the nominal size gap is the price of baseline over-coverage. We therefore run the coverage-matched ablation the previous draft deferred: holding GWD at $\alpha = 0.10$, we re-tune each baseline to a level α' at which its realized eval coverage matches GWD's on the same split, then compare (cx, cy)-slice areas (Table IX; frozen split and R=20 range, realized coverage matched to within ≤ 0.0002). Most of the nominal advantage does not survive. Against the like-for-like naive Bonferroni baseline — which, like GWD, certifies the full 5-DOF oriented box — GWD stays smaller on DIOR-R across all four cross-family detectors now certified there (§V-G): $\approx 0.79\times$ ORCNN, $\approx 0.83\times$ RTMDet, $\approx 0.76\times$ RoI Transformer, $\approx 0.85\times$ S2A-Net (range 0.76–0.85$\times$; smaller on all $20/20$ splits in every one of

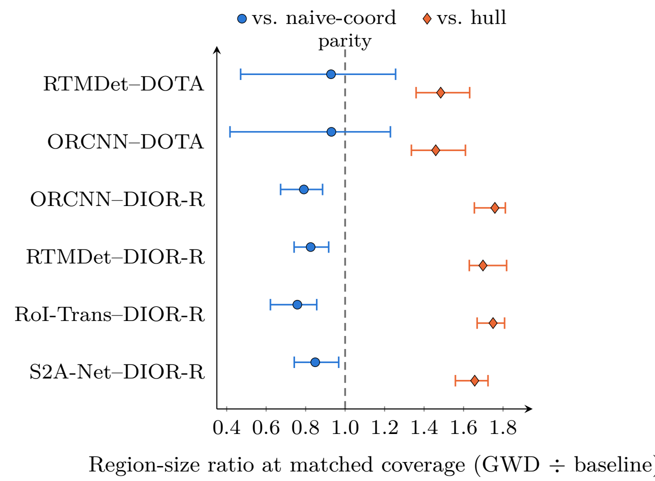

Fig. 3. Coverage-matched efficiency: per-cell region-size ratio (GWD $\div$ baseline; $<$1 favors GWD) against the naive coordinate-wise and axis-aligned-hull baselines, R=20 mean with $[\min, \max]$ range, at realized coverage matched to the GWD certificate. The dashed line marks parity (ratio = 1). All four DIOR-R cells sit below parity against the naive baseline; both DOTA cells' ranges straddle it; every cell sits above parity against the hull. From the frozen coverage-matched records (Table IX).

the four cells), but on DOTA it is a wash (mean ≈ 0.93, only $12/20$ splits favor GWD, the range spanning 1). Against the axis-aligned hull, GWD is larger at matched coverage in every cell run under this ablation, now six in total (1.46–1.76$\times$ the four original cells; 1.66$\times$ S2A-Net and 1.75$\times$ RoI Transformer, both within the same band): the nominal hull advantage was over-coverage plus the hull certifying only the coarser 4-DOF axis-aligned envelope (it drops the angle), so its center footprint is tighter at matched marginal coverage — a representational caveat, not a GWD defect. At the pooled level the coverage-fair advantage is narrow: GWD is $\approx 0.8\times$ the like-for-like naive oriented-box region on DIOR-R (all four detectors, every split), a wash on DOTA, and larger than the hull everywhere; but this pooled view averages over object shape, and conditioning on it (next, Table X; the regime-conditional analysis below is not re-run on the two Config-B DIOR-R cells) shows that, within each of the four frozen cells, the naive-baseline advantage is larger on the more elongated objects — a descriptive within-cell pattern, not a mechanism. This ablation is post-hoc but coverage-fair, and is not confirmatory (no preregistration claim). The premise-death branch of C2 did not fire, and the paper's correctness contributions (seam-continuity, square-safety, the coupled G1/G2 guarantees) do not rest on the efficiency margin. Figure 3 plots the per-cell coverage-matched ratio against both baselines directly, with the R=20 range and parity (ratio = 1) marked.

Within each frozen cell, the coverage-fair naive-baseline advantage is larger on more elongated objects (descriptive). The pooled ratios above average over object shape, which hides the mechanism. Repeating the coverage-matched comparison conditioned on the matched ground-truth box's aspect ratio $\text{AR} = \max(w, h)/\min(w, h)$ — a property of the ground-truth object, computed without reference to the certified region, so the strata are not circular, and reusing the parent's per-split

TABLE VIII

ACROSS-SPLIT SENSITIVITY (R=20, NOT THE PREREGISTERED HOLM-8) OF THE GWD-VS-BASELINE REGION-SIZE RATIO AT MATCHED NOMINAL α (GWD $\div$ BASELINE; <1 FAVORS GWD). PER CELL: MEAN AND $[\min, \max]$ OVER 20 SCENE SPLITS (SPLIT-SEED = REPEAT INDEX, SAME PROTOCOL AS THE G1 HEADLINE) AND THE COUNT OF SPLITS WITH GWD SMALLER. THE 20 SPLITS RESHUFFLE ONE FIXED SCENE SET (DEPENDENT), SO NO p-VALUE IS ATTACHED; EVERY SPLIT FAVORS GWD AND NO RATIO CROSSES 1. THESE NOMINAL-LEVEL RATIOS ARE RE-ASSESSED ON A COVERAGE-FAIR FOOTING IN TABLE IX.

Cell	Baseline	mean ratio	$[\min, \max]$	GWD smaller
RTMDet–DOTA	naive	0.597	$[0.519, 0.791]$	$20/20$
RTMDet–DOTA	hull	0.773	$[0.687, 0.974]$	$20/20$
ORCNN–DOTA	naive	0.573	$[0.468, 0.683]$	$20/20$
ORCNN–DOTA	hull	0.746	$[0.635, 0.876]$	$20/20$
ORCNN–DIOR-R	naive	0.342	$[0.307, 0.395]$	$20/20$
ORCNN–DIOR-R	hull	0.493	$[0.460, 0.567]$	$20/20$
RTMDet–DIOR-R	naive	0.381	$[0.358, 0.431]$	$20/20$
RTMDet–DIOR-R	hull	0.549	$[0.505, 0.625]$	$20/20$

TABLE IX

COVERAGE-MATCHED EFFICIENCY ABLATION (POST-HOC, COVERAGE-FAIR; NOT PREREGISTERED). REGION-SIZE RATIO (GWD $\div$ BASELINE; <1 FAVORS GWD) WITH EACH BASELINE RE-TUNED TO GWD'S REALIZED EVAL COVERAGE. COLUMNS: THE NOMINAL-α RATIO (TABLE VIII FOR THE FOUR ORIGINAL CELLS; FOR THE TWO CONFIG-B DIOR-R CELLS, NOT PART OF THAT PREREGISTERED FAMILY, THIS ABLATION'S OWN R=20 NOMINAL-α MEAN) AND THE COVERAGE-MATCHED RATIO (R=20 MEAN $[\min, \max]$), PLUS SPLITS WITH GWD SMALLER. THE NAIVE BASELINE CERTIFIES THE FULL ORIENTED BOX (LIKE GWD); THE HULL CERTIFIES ONLY THE COARSER AXIS-ALIGNED ENVELOPE. FROM THE COVERAGE-MATCHED RECORDS.

Cell	Baseline	nominal	matched $[\min, \max]$	GWD smaller
RTMDet–DOTA	naive	0.60	0.93 $[0.47, 1.26]$	$12/20$
RTMDet–DOTA	hull	0.77	1.48 $[1.36, 1.63]$	$0/20$
ORCNN–DOTA	naive	0.57	0.93 $[0.42, 1.23]$	$12/20$
ORCNN–DOTA	hull	0.75	1.46 $[1.34, 1.61]$	$0/20$
ORCNN–DIOR-R	naive	0.34	0.79 $[0.67, 0.89]$	$20/20$
ORCNN–DIOR-R	hull	0.49	1.76 $[1.65, 1.81]$	$0/20$
RTMDet–DIOR-R	naive	0.38	0.83 $[0.74, 0.92]$	$20/20$
RTMDet–DIOR-R	hull	0.55	1.70 $[1.63, 1.82]$	$0/20$
RoI-Trans–DIOR-R	naive	0.33	0.76 $[0.62, 0.86]$	$20/20$
RoI-Trans–DIOR-R	hull	0.47	1.75 $[1.67, 1.81]$	$0/20$
S2A-Net–DIOR-R	naive	0.38	0.85 $[0.74, 0.97]$	$20/20$
S2A-Net–DIOR-R	hull	0.54	1.66 $[1.56, 1.72]$	$0/20$

calibration verbatim (the unconditioned "all" regime reproduces Table IX bit-for-bit; regression-guarded) — resolves the pooled numbers into a monotone within-cell trend across aspect-ratio strata (Table X). Against the like-for-like naive per-coordinate baseline, at realized coverage matched to ≤ 0.0004 in every regime, GWD's (cx, cy) region is smaller for elongated objects ($\mathrm{AR} \geq 3$) in all four original cells (0.74–0.87×), while for compact objects ($\mathrm{AR} < 3$) it is a wash-to-slight-loss (0.97–1.08×, the slight loss confined to DOTA). A per-cell median split (balanced halves, cut $\approx$ 2.2–2.5) gives the same direction in all four original cells and sharpens it on DIOR-R, where the more-elongated half reaches 0.59–0.65× against $\approx 1.02\times$ on the compact half. This explains the DOTA "wash" (≈ 0.93) rather than explaining it away: it is the average of a genuine elongated-object win and a slight compact-object loss, not evidence that the angle-aware region is never sharper — within these four frozen cells the efficiency advantage is concentrated on the more elongated objects. We report this as a description of these four cells, not as a mechanism: HRSC2016-MS, an all-elongated population (eval frac $\mathrm{AR} \geq 3$ 0.90/0.93), landed at pooled coverage-matched ratios 1.2413/1.0198 — outside the four cells' elongated band (0.74–0.87×) and in the wrong direction — so the within-cell pattern is descriptive only and did not survive its one out-of-sample test (§V-F).

Two scope limits bound the claim honestly. First, this is specifically a repair versus the naive per-coordinate baseline: against the axis-aligned hull GWD is larger in every regime (1.4–1.8×) with no elongation gradient (elongated $\geq$ compact in $3/4$ cells), so the advantage is not "smaller than the hull." Second, the between-dataset pattern is not an elongation effect: DOTA objects are on average more elongated than DIOR-R (mean AR 2.70 vs. 2.35), yet GWD wins pooled on DIOR-R and only ties on DOTA, so the DIOR-vs-DOTA gap tracks the DOTA crop/exchangeability artifact (§V), not shape — the elongation effect is within-cell and conditional. Finally, as a registered prediction (logged 2026-07-13, before the run) for the staged third-dataset arm: HRSC2016-MS ships (single-class; mean AR 5.22, 82% at $\mathrm{AR} \geq 3$) lie almost entirely in the elongated regime, so their pooled coverage-matched ratio versus naive-coord is predicted to be the most decisive of any cell (low-0.7× or below); a wash there would falsify the elongation mechanism. This prediction was executed and reversed (§V-F): HRSC pooled ratios 1.2413/1.0198 are both ≥ 1.0, so the elongation mechanism does not transfer across datasets and no cross-dataset efficiency claim is made.

Confinement of the coverage-fair efficiency claim. Netted across datasets, the coverage-fair naive-baseline size advantage is currently demonstrated on one of the three: GWD is ≈ 0.76–$0.85\times$ the like-for-like naive region on DIOR-R (all four detectors — ORCNN, RTMDet, RoI Transformer, S2A-Net — all $20/20$ splits), a wash on DOTA (≈ 0.93), and reversed on HRSC2016-MS ($1.2413/1.0198$). The DOTA wash was attributed to a crop/exchangeability artifact of DOTA's overlapping tiles, but that account cannot rescue the claim's generality: HRSC2016-MS is certified single-tile (scene $=$ image, uncropped, 453 whole test images resized in place), so its reversal on the most elongated dataset cannot be a crop artifact. The efficiency advantage is therefore reported as DIOR-R-specific, not general.

VI. Limitations

Certificate scope (v1). G1 is conditional on detection: it certifies localization for matched true positives, not whether an object is detected (G2's separate guarantee). The certified object is the GWD-ball in Gaussian-Wasserstein space, not an IoU level-set (Methods); its per-parameter envelope over-covers and is not a set of calibrated coordinate intervals. Guarantees are valid for the calibration distribution only; the tool prints a recalibrate-on-drift caveat.

Scale and the DOTA under-coverage. The out-of-sample DOTA coverage (0.888) sits ≈ 1.2 points below nominal; we attribute this to residual within-scene correlation in DOTA's overlapping crops (the same detector covers at nominal on single-tile DIOR-R), but a weighted/adaptive conformal remedy for the crop structure is future work. The $R{=}20$ per-split spread bounds how much this varies by realization.

Frozen-checkpoint variance. The DOTA detector is a frozen published checkpoint (no training-seed variance; variance is over split repeats). The DIOR-R and HRSC2016-MS detectors are trained in house; for HRSC2016-MS a post-hoc 3-seed arm (Table VII) quantifies training-seed variance: audited coverage is seed-stable by construction (within-detector spread ≤ 0.0013) while the certified region scale $\hat{q}$ does vary with seed (up to $\approx 12\%$ for RTMDet-R), and all four DIOR-R cells remain single-seed (multi-seed DIOR-R detector variance is future work). The two DOTA zoo detectors (RoI Transformer, Oriented RepPoints, §V-F) are frozen published checkpoints; note that "RoI Transformer" therefore names two distinct trained instances in this paper — a frozen DOTA-v1.0 zoo checkpoint (§V-F) and an independently in-house-trained DIOR-R model (§V-G) — never the same weights certified twice.

Statistical significance and scoop currency. The regime-conditional under-coverage contrast is descriptive here; the preregistered Holm set-size head-to-head ran under the signed freeze and is reported in §V-H, where the frozen single-split family is demoted to a preregistered descriptive outcome (GWD smaller in $8/8$ cells at nominal α) — structurally degenerate, so not presented as confirmatory and not amended. Because that $8/8$ is at matched nominal α, where the baselines over-cover, our primary quantitative characterization of efficiency is the post-hoc, coverage-fair coverage-matched ablation (§V-H, Tables IX–X): it shows the size advantage is largely baseline over-coverage, and that the residual advantage over the like-for-like naive baseline is, descriptively and within each of the four original cells, larger on more elongated objects — GWD's region is smaller for elongated objects (aspect ratio ≥ 3) in all four original cells (0.74–$0.87\times$) and a wash-to-slight-loss for compact ones — while against the axis-aligned hull GWD is larger in every regime. This gradient is now scoped to a within-cell, descriptive observation: the falsifiable prediction for the elongation-dominated HRSC2016-MS ship arm was executed and reversed (§V-F; HRSC pooled ratios $1.2413/1.0198$, both ≥ 1.0), so the elongation mechanism does not transfer across datasets or detectors. The $R{=}20$ across-split analysis is reported as split-sensitivity of the nominal contrast, with no p-value (the 20 splits reshuffle one fixed scene set and are not independent). The differentiation against EAV-DETR (22) (§II) rests on its public abstract and code release, its full text being paywalled. The DIOR-R AOPG mAP reproduction gate was executed (reproduced test mAP 62.61 / 68.36 vs. AOPG 64.41; ± 0.5 not met cross-method; same-method external anchor 64.30 (40), disclosed).

VII. Full-study design

The pilot and DIOR-R cells instantiate the G1/G2 arms; the preregistered study adds the $R{=}20$ repeats for every cell (done for the coverage headline) and the preregistered Holm family (the paper's only Holm family: 2 baseline contrasts $\times$ 2 detectors $\times$ 2 datasets $=$ 8 tests), which ran under the signed freeze and is reported in §V-H (GWD smaller in $8/8$ cells, demoted to a preregistered descriptive outcome; no amendment). The AOPG mAP reproduction gate has now been executed (§V); no preregistration-gated slot remains. The staged third-dataset (HRSC2016-MS) arm and the DOTA detector-zoo breadth were executed as the Config-A extension (§V-F); the registered elongation prediction it tested is reported reversed (§V-H). The cross-family DIOR-R detector-breadth arm (RoI Transformer, S2A-Net) was executed as the Config-B extension (§V-G); both new cells certify at G1 with zero Mondrian refusals.

Data and Code Availability

DOTA-v1.0 (public; val GT used, test server untouched) and DIOR-R (public; in-house-trained detectors under Apache-2.0 mmrotate). The `rotcert` tool (numpy/scipy/shapely only; no torch/mmrotate in the certifier) and all frozen result records (DOTA pilot, DIOR-R Oriented R-CNN and RTMDet-R-l, the out-of-sample $R{=}20$ coverage, the naive regime audits, the frozen preregistered head-to-head record, the across-split analysis record, and the coverage-matched ablation record) accompany the paper; the CLI is golden-file-tested against these records. The `rotcert` tool and all frozen result records are released as a public GitHub repository, archived at a versioned Zenodo DOI on acceptance (repository URL and DOI: TODO-USER). Per the target venues' live policies (verified 2026-07-15), GitHub plus a Zenodo DOI satisfies both the Elsevier data-availability mechanism (ISPRS J. P&RS) and IEEE's optional reproducibility program (Code Ocean remains opt-in and is not used here). Provenance note: the frozen head-to-head record was written with a dirty working tree (uncommitted changes present when the preregistered family ran); the exact script and environment are stamped in the record's own provenance

TABLE X

REGIME-CONDITIONAL COVERAGE-MATCHED EFFICIENCY VERSUS THE NAIVE PER-COORDINATE BONFERRONI BASELINE (POST-HOC, COVERAGE-FAIR; NOT PREREGISTERED). RATIO GWD ÷ NAIVE OF THE (cx, cy)-SLICE AREA (<1 FAVORS GWD), CONDITIONED ON THE MATCHED GROUND-TRUTH BOX ASPECT RATIO $\mathrm{AR} = \max(w, h)/\min(w, h)$; REALIZED COVERAGE MATCHED TO ≤ 0.0004 IN EVERY REGIME. FOR THE FOUR ORIGINAL FROZEN CELLS THE "ALL" COLUMN REPRODUCES TABLE IX'S POOLED RATIO BIT-FOR-BIT (REGRESSION-GUARDED), AND ELONGATED OBJECTS FAVOR GWD WHILE COMPACT OBJECTS ARE A WASH-TO-SLIGHT-LOSS. THE OUT-OF-SAMPLE CONFIG-A EXTENSION CELLS (HRSC2016-MS, DOTA-V1.0 ZOO; §V-F) DO NOT TRANSFER THE GRADIENT — HRSC IS ≥ 1 DESPITE BEING ALMOST ALL ELONGATED, AND THE DOTA-ZOO CELLS ARE NEAR-ALL COMPACT (—=REGIME WITH NO POWERED SPLIT). AGAINST THE AXIS-ALIGNED HULL THE GRADIENT DOES NOT HOLD (TEXT). FROM THE REGIME-CONDITIONAL COVERAGE-MATCHED RECORDS.

Cell	AR mean (frac ≥3)	all	compact (AR<3)	elongated (AR≥3)
Four original frozen cells (within-cell elongation gradient)				
RTMDet–DOTA	2.69 (0.29)	0.93	1.03	0.85
ORCNN–DOTA	2.71 (0.30)	0.93	1.08	0.74
ORCNN–DIOR-R	2.38 (0.22)	0.79	0.98	0.87
RTMDet–DIOR-R	2.34 (0.21)	0.83	0.97	0.86
Out-of-sample Config-A extension (gradient does not transfer)				
HRSC ORCNN	5.63 (0.90)	1.24	—	1.26
HRSC RTMDet-R	5.95 (0.93)	1.02	—	0.98
DOTA RoI Transf.	1.27 (0.00)	1.66	1.66	—
DOTA RepPoints	1.47 (0.04)	0.71	0.74	—

metadata. Detector weights are frozen published checkpoints (DOTA) or in-house-trained (DIOR-R, redistribution subject to the DIOR-R license review; result-records-only if rehosting is barred). The Config-A extension records (four new G1 certification cells on HRSC2016-MS and the DOTA detector zoo, their R=20 recalibration, the G2 power-floor refusals, and the reversed registered-prediction verdict) accompany the paper as well, as do the Config-B extension records (two new G1 certification cells on DIOR-R — RoI Transformer, S2A-Net — their R=20 recalibration, G2 status, and the extended four-cell per-class conditional-coverage record).

REFERENCES

[1] C. Lyu, W. Zhang, H. Huang, Y. Zhou, Y. Wang, Y. Liu, S. Zhang, and K. Chen, "RTMDet: An empirical study of designing real-time object detectors," *arXiv preprint arXiv:2212.07784*, 2022.

[2] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2021, arXiv:2108.05699.

[3] Y. Li, Q. Hou, Z. Zheng, M.-M. Cheng, J. Yang, and X. Li, "Large selective kernel network for remote sensing object detection," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2023, arXiv:2303.09030.

[4] G.-S. Xia, X. Bai, J. Ding, Z. Zhu, S. Belongie, J. Luo, M. Datcu, M. Pelillo, and L. Zhang, "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2018, arXiv:1711.10398.

[5] G. Cheng, J. Wang, K. Li, X. Xie, C. Lang, Y. Yao, and J. Han, "Anchor-free oriented proposal generator for object detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, 2022, arXiv:2110.01931; DOI 10.1109/TGRS.2022.3183022.

[6] L. Andéol, T. Fel, F. De Grancey, and L. Mossina, "Confident object detection via conformal prediction and conformal risk control: an application to railway signaling," *arXiv preprint arXiv:2304.06052*, 2023.

[7] A. Timans, C.-N. Straehle, K. Sakmann, and E. Nalisnick, "Adaptive bounding box uncertainties via two-step conformal prediction," in *European Conference on Computer Vision (ECCV)*, 2024, arXiv:2403.07263.

[8] L. Andéol, L. Mossina, A. Mazoyer, and S. Gerchinovitz, "Conformal object detection by sequential risk control," *arXiv preprint arXiv:2505.24038*, 2025.

[9] C. Ries, M. Kassem Sbeyti, N. Bianco, and N. Klein, "Probabilistic object detection with conformal prediction," *arXiv preprint arXiv:2605.07549*, 2026.

[10] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," *arXiv preprint arXiv:2208.02814*, 2022.

[11] V. Copley, G. Finlay, and B. Hiett, "The uncertain object: Application of conformal prediction to aerial and satellite images," in *Proc. 13th Symp. Conformal and Probabilistic Prediction with Applications (COPA), PMLR*, vol. 230, 2024, pp. 73–89.

[12] X. Yang, J. Yan, Q. Ming, W. Wang, X. Zhang, and Q. Tian, "Rethinking rotated object detection with Gaussian Wasserstein distance loss," in *Proc. Int. Conf. Machine Learning (ICML)*, 2021, arXiv:2101.11952.

[13] X. Yang, X. Yang, J. Yang, Q. Ming, W. Wang, Q. Tian, and J. Yan, "Learning high-precision bounding box for rotated object detection via Kullback-Leibler divergence," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2021, arXiv:2106.01883.

[14] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic Learning in a Random World*. Springer, 2005.

[15] A. N. Angelopoulos, S. Bates, E. J. Candès, M. I. Jordan, and L. Lei, "Learn then test: Calibrating predictive algorithms to achieve risk control," *arXiv preprint arXiv:2110.01052*, 2021.

[16] J. Ding, N. Xue, Y. Long, G.-S. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2019, pp. 2849–2858.

[17] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features

for oriented object detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 60, pp. 1–11, 2022, arXiv:2008.09397; DOI 10.1109/TGRS.2021.3062048.

[18] V. Vovk, D. Lindsay, I. Nouretdinov, and A. Gammerman, "Mondrian confidence machine," Royal Holloway, University of London, Technical Report, 2003.

[19] Y. Romano, E. Patterson, and E. J. Candès, "Conformalized quantile regression," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019, pp. 3538–3548.

[20] S. Bates, A. Angelopoulos, L. Lei, J. Malik, and M. I. Jordan, "Distribution-free, risk-controlling prediction sets," *Journal of the ACM*, vol. 68, no. 6, pp. 1–34, 2021, arXiv:2101.02703.

[21] A. N. Angelopoulos, A. P. Kohli, S. Bates, M. Jordan, J. Malik, T. Alshaabi, S. Upadhyayula, and Y. Romano, "Image-to-image regression with distribution-free uncertainty quantification and applications in imaging," in *Proc. Int. Conf. Machine Learning (ICML)*, 2022, pMLR 162:717–730; arXiv:2202.05265.

[22] H. Zuo, M. Ning, Y. Shu, S. Huang, and C. Sun, "EAV-DETR: Efficient arbitrary-view oriented object detection with probabilistic guarantees for UAV imagery," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 233, pp. 575–587, 2026, dOI 10.1016/j.isprsjprs.2026.02.009; code github.com/zzzhak/EAV-DETR.

[23] J. Li, W. He, W. Cao, L. Zhang, and H. Zhang, "UANet: An uncertainty-aware network for building extraction from remote sensing images," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, 2024, art. 6261211; DOI 10.1109/TGRS.2024.3361211.

[24] F. Ji, W. Zhao, Q. Wang, W. J. Emery, R. Peng, Y. Man, G. Wang, and K. Jia, "Spectral-spatial evidential learning network for open-set hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, 2024, dOI 10.1109/TGRS.2024.3349415.

[25] Z. Huang, Y. Liu, X. Yao, J. Ren, and J. Han, "Uncertainty exploration: Toward explainable SAR target detection," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1–14, 2023, dOI 10.1109/TGRS.2023.3247898.

[26] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. Int. Conf. Machine Learning (ICML)*, 2017, pp. 1321–1330, pMLR 70:1321–1330; arXiv:1706.04599.

[27] F. Küppers, J. Kronenberger, A. Shantia, and A. Haselhoff, "Multivariate confidence calibration for object detection," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2020.

[28] M. A. Munir, M. H. Khan, M. S. Sarfraz, and M. Ali, "Towards improving calibration in object detection under domain shift," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2022, arXiv:2209.07601.

[29] J. Ding, N. Xue, G.-S. Xia, X. Bai, W. Yang, M. Y. Yang, S. Belongie, J. Luo, M. Datcu, M. Pelillo, and L. Zhang, "Object detection in aerial images: A large-scale benchmark and challenges," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2021, arXiv:2102.12219.

[30] Z. Liu, L. Yuan, L. Weng, and Y. Yang, "A high resolution optical satellite image dataset for ship recognition and some new baselines," in *Proc. 6th Int. Conf. Pattern Recognition Applications and Methods (ICPRAM)*, 2017, pp. 324–331.

[31] X. Sun, P. Wang, Z. Yan, F. Xu, R. Wang, W. Diao, J. Chen, J. Li, Y. Feng, T. Xu, M. Weinmann, S. Hinz, C. Wang, and K. Fu, "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 184, pp. 116–130, 2022, arXiv:2103.05569; DOI 10.1016/j.isprsjprs.2021.12.004.

[32] W. Li, Y. Chen, K. Hu, and J. Zhu, "Oriented RepPoints for aerial object detection," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022, arXiv:2105.11111.

[33] J. Ma, W. Shao, H. Ye, L. Wang, H. Wang, Y. Zheng, and X. Xue, "Arbitrary-oriented scene text detection via rotation proposals," *IEEE Transactions on Multimedia*, vol. 20, no. 11, pp. 3111–3122, 2018, dOI 10.1109/TMM.2018.2818020.

[34] Y. Xu, M. Fu, Q. Wang, Y. Wang, K. Chen, G.-S. Xia, and X. Bai, "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 4, pp. 1452–1459, 2021, arXiv:1911.09358; DOI 10.1109/TPAMI.2020.2974745.

[35] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artificial Intelligence*, vol. 35, no. 4, 2021, pp. 3163–3171, arXiv:1908.05612.

[36] J. Han, J. Ding, N. Xue, and G.-S. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2021, arXiv:2103.07733.

[37] X. Yang, J. Yang, J. Yan, Y. Zhang, T. Zhang, Z. Guo, X. Sun, and K. Fu, "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2019, pp. 8232–8241, arXiv:1811.07126.

[38] Y. Zhou, X. Yang, G. Zhang, J. Wang, Y. Liu, L. Hou, X. Jiang, X. Liu, J. Yan, C. Lyu, W. Zhang, and K. Chen, "MMRotate: A rotated object detection benchmark using PyTorch," in *Proc. 30th ACM Int. Conf. Multimedia (MM)*, 2022, arXiv:2204.13317.

[39] F. de Grancey, J.-L. Adam, L. Alecu, S. Gerchinovitz, F. Mamalet, and D. Vigouroux, "Object detection with probabilistic guarantees: A conformal prediction approach," in *Computer Safety, Reliability, and Security. SAFECOMP 2022 Workshops (WAISE)*, ser. Lecture Notes in Computer Science. Springer, 2022, dOI 10.1007/978-3-031-14862-0_23.

[40] Z. Ding, Y. Zhou, J. Zhao, W.-L. Du, X. Li, R. Yao, and A. El Saddik, "Real-time oriented object detection transformer in remote sensing images," *arXiv preprint arXiv:2603.15497*, 2026.